MIL-S-46070(MU)
AMENDMENT 2
17 January 1975
SUPERSEDING
AMENDMENT 1
26 September 1969

MILITARY SPECIFICATION

STEEL BAR FOR PROJECTILE STOCK

This amendment forms a part of Military Specification MIL-S-46070(MU), 12 February 1968, and is approved for use by all Departments and Agencies of the Department of Defense.

Para 2.1, under STANDARDS, delete MILITARY, MIL-STD-430A - Macrograph Standards for Steel Bars, Billets and Blooms.

Para 2.2 Other Publications, add the following,

"ASTM-E381 - Standard Method of Rating Macroetched Steel"

Para 3.3, delete and substitute the following:

"3.3 Chemical composition.-The heat analysis of each heat shall meet the chemical composition shown in Table I for the grade ordered. (see 6.2)

Table I

Grade	Carbon Percent	Manganese Percent	Phosphorus Percent	Sulfur Percent	Silicon Percent
1030 to 1040	0.28 to 0.44	0.60 to 0.90	0.035 max	0.045 max	0.30 max
			"	"	"

Incidental elements, max. percent Cr 0.20, Cu 0.35, Ni 0.25 and Mo 0.06.

The chemical composition of billets or bars, as determined by product analysis, shall meet the chemical requirements shown in Table I within the permissible variations shown in Table II, provided that in each heat of steel the percentage of any element shall not vary both above and below the range shown in Table I.

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Table II

Permissible Variations in Chemical Composition for Product Analysis

Carbon Percent		Manganese Percent		Phosphorus Percent	Sulphur Percent	Silicon Percent
Over	Under	Over	Under	Over	Over	Over
0.03	0.03	0.03	0.03	0.008	0.008	0.02"

Para 3.6.1.1, delete and substitute the following:

"3.6.1.1 Billet.-A macroscopic examination of billets shall show no unsound conditions including refractory materials of a nature, degree or extent in excess of the following standard ratings: C-3, R-4, S-4 of ASTM-E-381."

Para 3.6.1.2, delete and substitute the following:

"3.6.1.2 Bar.-A macroscopic examination of finished bars shall show no unsound condition including refractory materials of a nature, degree or extent in excess of the following standard ratings: C-3, R-4, S-4 of ASTM-E-381. This requirement is not applicable when the steel has met the requirements of 3.6.1.1."

Para 3.6.2, delete and substitute the following:

"3.6.2 Magnetic.-Finished bars shall be free of cracks, seams, folds, laps, scabs, or slivers which extend to a depth greater than 0.020 inch below the surface of the bar, except on bars over 1½ inches in diameter where defects shall be limited to 0.001 inch in depth per 1/16 inch in bar diameter."

Para 4.5.4.2, delete and substitute the following:

"4.5.4.2 Magnetic.-The test system shall incorporate scanning rotation, an indicating or rejection device and shall operate at a frequency not exceeding 75 kilohertz (kHz). The rejection standard for calibration control shall have slots 1.250 inches maximum in length, 0.010 inch maximum in width and 0.018 inch maximum in depth "

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Add the following paragraph after para 4.5.6:

"4.6 Magnetic inspection system.-The electromagnetic inspection system utilized shall conform to the requirements set forth in this specification. A description of the inspection equipment and operating procedures shall be supplied to the cognizant technical agency (see 6.5) for review and retention. Inadequacies in the system shall be corrected as specified by the Government. The system shall be subject to inspection at the convenience of the Government."

Add the following paragraph after para 6.4:

"6.5 Magnetic inspection system review.-Submission of inspection procedures to establish compliance with 4.6 may be made by potential suppliers of steel bar by mailing to Commander, Frankford Arsenal, ATTN: SARFA-QAA, Philadelphia, Pa. 19137."

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